Measuring School Security: A Rasch Modeling Approach Using ELS:2002

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1 Introduction

Schools have a security problem. Not always a crime problem: a security *measurement* problem. Concepts like "security climate" are not directly observable. Administrators implement dozens of different measures simultaneously; no single item captures the whole picture. Combining them requires a principled model.

Finn and Servoss (2014) addressed this by fitting a Rasch model to seven binary items from the Education Longitudinal Study of 2002 (ELS:2002), producing a single latent score per school. Their index has been used to study the relationship between security practices, race, and student discipline. A team of researchers now wants to build on this work using more recent data. Before they do, they need to know: is the F&S measurement model worth keeping, or can we do better?

I was asked to: (a) replicate the F&S Rasch model and fit a penalized variant; (b) construct and evaluate an alternative model using different items; (c) compare the two models statistically; (d) examine how similarly the two indices rank schools; and (e) test whether security differs across school types, examining all four available school-level characteristics with appropriate multiplicity correction. My short answer: the F&S model fits better, but item choice matters more than I expected. The two indices disagree sharply on school ordering, and that finding has real policy consequences.

2 Methods

I used two files from ELS:2002. The primary dataset, `security_wide.csv`, contains school-level binary responses to 20 administrator survey items on security practices. A supplementary file, `security_school_vars.csv`, provides four school characteristics: urbanicity (`BYURBAN`, 3 levels), 10th-grade enrollment size (`BYG10EP`), geographic region (`BYREGION`), and the percentage of 10th-grade students eligible for free or reduced-price lunch (`BY10FLP`) (National Center for Education Statistics, 2004). Values of -9 indicate missing data; I recoded them as `NA` before merging.

The Rasch model treats each school's observed response as arising from a latent security orientation. Let θ_i denote school i's level and δ_j the rarity of item j across schools:

$$\Pr(Y_{ij} = 1) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)} \tag{1}$$

Schools with higher θ_i implement more measures. I fit the model using `lme4::glmer()` (Bates et al., 2015) with school effects as random intercepts and item effects as fixed intercepts, binomial family, logit link. Schools with partial response patterns are included under a **missing at random (MAR)** assumption: whether a school skipped an item may depend on observed characteristics, but not on its unobserved security level. I fit a penalized variant by treating item effects as random intercepts instead of fixed, shrinking difficulty estimates toward zero. With rare items, maximum likelihood assigns extreme difficulty estimates that distort school scores at the tails; penalization trades a small amount of in-sample fit for stability. I compared all four models on average deviance (deviance divided by the number of school-item observations) and AIC per observation (AIC/N), both of which allow comparison across models with different observation counts.

The F&S item set (`q38a`–`q38g`) captures metal detectors, security guards, locked building doors, visitor sign-in requirements, random locker and bag checks, required student ID badges, and drug testing. My alternative set (`q38h`–`q38n`) captures surveillance cameras, controlled building entry, permanently assigned school resource officers, random searches with detection dogs, dress codes, closed-campus policies, and required clear backpacks. These target practices students encounter daily; a student may never see a metal detector but passes through a camera or locked entry every day. The two sets are non-overlapping.

To test whether school characteristics predict security, I selected all four available school-level variables as my planned hypothesis set: urbanicity, region, enrollment size, and lunch eligibility. I had prior reasons to expect each: region and urbanicity reflect contextual differences in crime and community norms; size and economic disadvantage have been linked to perceived disorder and institutional control in prior research. For each variable, I compared two nested `glmer()` models on the same observations — one without the variable, one with it as a fixed effect — using a likelihood ratio test (LRT). The family of hypotheses covers only whether each variable matters as a whole. Pairwise contrasts between individual levels are treated as exploratory follow-ups and are not subject to the primary correction; this keeps $k = 4$ and preserves more power than expanding the family to all pairwise contrasts. I selected Bonferroni as my multiplicity correction: it is straightforward to implement, does not require assumptions about dependence among tests, and is conservative in a transparently defensible way. The corrected threshold is $\alpha^* = 0.05/4 = 0.0125$.

3 Results

Both item sets drew on 656 schools with at least one non-missing response and 638 with complete data across all seven items (Table 1). Missingness patterns are identical across the two item sets; item choice does not affect sample size here.

Table 1: Schools contributing to each model.

Model	Schools (any response)	Schools (complete)
F&S (`q38a--q38g`)	656	638
Alternative (`q38h--q38n`)	656	638

Table 2 shows average deviance and AIC/N for all four fitted models. The F&S ML model achieves the lowest average deviance (0.841). The alternative fits noticeably worse (0.910). Penalization worsens in-sample fit slightly for both, as expected: shrinking item difficulty estimates toward zero reduces variance at the cost of bias, so deviance rises. AIC is not reported for penalized models; treating items as random effects changes the likelihood structure in a way that makes AIC non-comparable to the fixed-effects versions. On fit alone, I prefer the F&S item set. The 0.069-unit gap in average deviance is not trivial.

Table 2: Model fit comparison. Average deviance (deviance/N) and AIC/N allow comparison across models with different observation counts. Lower = better. AIC undefined for penalized models; see text.

Model	N (school-items)	Avg. Deviance	AIC/N
F&S (ML)	4,563	0.8409	0.9986
F&S (Penalized)	4,563	0.8463	—
Alternative (ML)	4,566	0.9101	1.0445
Alternative (Penalized)	4,566	0.9145	—

The Spearman rank correlation between the F&S and alternative security indices is 0.31; Kendall's $\tau = 0.225$. Both are low. Two indices measuring the same underlying trait should agree substantially on how they order schools. They do not here. Figure 1 makes this plain: the scatter shows no clean trend; schools land across the entire plot with no systematic pattern. Figure 2 shows the rank difference distribution — wide, roughly symmetric, spanning from -600 to $+600$. Many schools shift by more than 200 positions depending on which items are used. This is the finding I take most seriously. The F&S model fits better statistically, but identifying a school as high-security depends heavily on which practices get asked about. There is no stable answer.

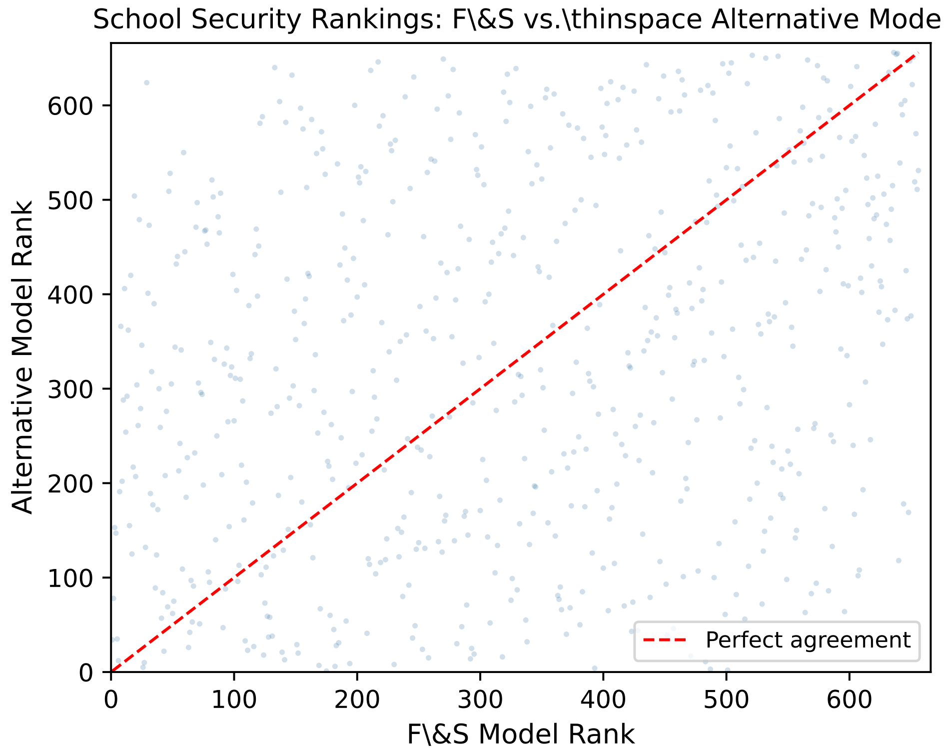

Figure 1: Each point is a school. Rank under the F&S model (horizontal) versus rank under the alternative model (vertical). The dashed red line shows perfect agreement. Spearman $\rho = 0.31$.

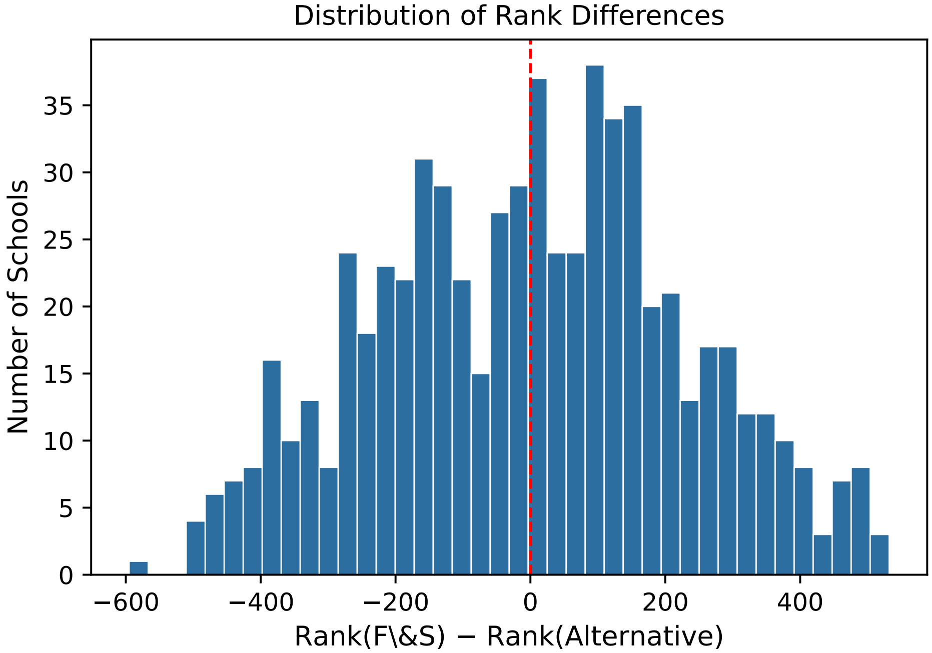

Figure 2: Distribution of rank differences (F&S rank minus alternative rank) across 656 schools. A narrow peak near zero would indicate agreement; the wide spread here confirms the indices order schools very differently.

Table 3 shows LRT results for all four school-level variables. Three are significant at $\alpha^* = 0.0125$: geographic region, enrollment size, and lunch eligibility. Urbanicity is not significant (corrected $p = 1.0$). The lunch eligibility test covers 603 schools rather than 656 due to missingness in that variable; both null and alternative models for that test were fit on the same 603 observations, so the comparison remains valid.

Table 3: Likelihood ratio tests for each school-level variable. Bonferroni correction: $k = 4$ planned tests, $\alpha^* = 0.0125$.

Variable	N Schools	Raw p	Bonf. p	Reject H_0?
Urbanicity	656	0.818	1.000	No
Region	656	<0.001	<0.001	Yes
Size (10th grade)	656	<0.001	<0.001	Yes
Lunch eligibility (%)	603	<0.001	<0.001	Yes

Table 4 shows the region coefficients. Regions 2 and 3 score substantially higher than the reference region ($\hat{\beta} = 0.523$ and 0.839 respectively); region 4 is nearly indistinguishable from it ($\hat{\beta} = -0.055$). The direction for size and lunch eligibility is positive in both cases: larger schools implement more intensive security, as do schools with higher proportions of economically disadvantaged students. Figure 3 shows security scores by urbanicity category; the three groups have nearly identical medians and heavily overlapping distributions, consistent with the non-significant LRT.

Table 4: Region coefficients from the augmented Rasch model. Reference: region 1. Positive estimates indicate higher security than the reference. Per-coefficient p-values not reported; the Bonferroni-corrected LRT p-value (< 0.001) applies to the variable as a whole.

Region	Estimate	Std. Error	z value
2	0.523	0.155	3.371
3	0.839	0.146	5.748
4	−0.055	0.167	−0.327

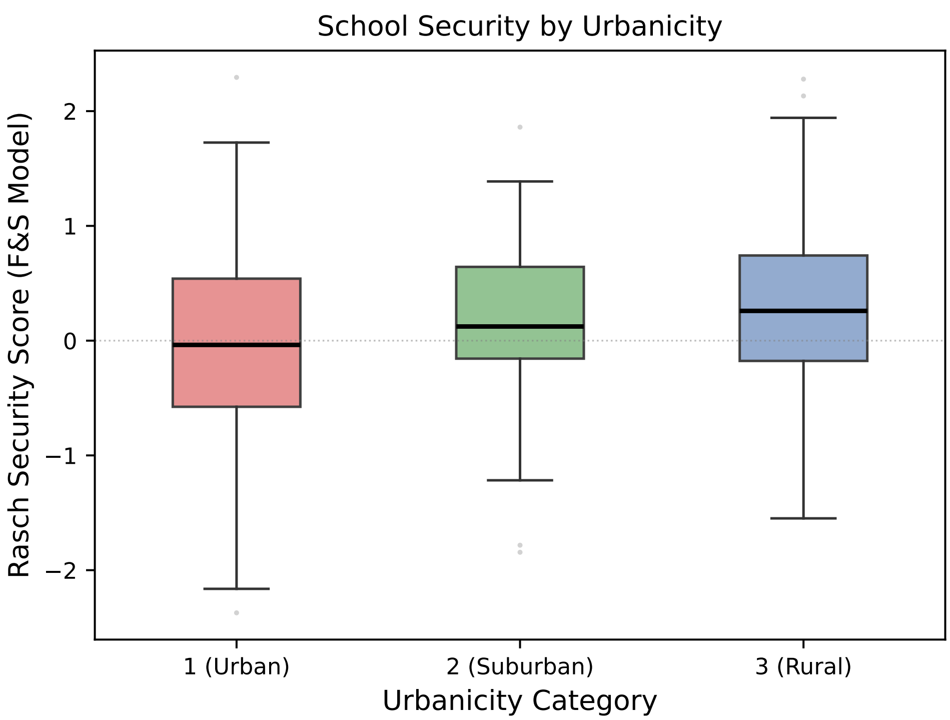

Figure 3: F&S Rasch security scores by urbanicity category (3 levels). The flat pattern is consistent with the non-significant LRT result (Bonferroni-corrected $p = 1.0$).

4 Discussion

I recommend the F&S item set with penalized fitting for the new study. The F&S ML model outperforms the alternative on average deviance by a margin large enough to matter (0.841 vs. 0.910). Whatever the alternative items are capturing, it is noisier. Penalized fitting is preferable to plain maximum likelihood because some items are rare; rare items produce extreme estimated difficulty parameters that distort school scores at the tails. Shrinking these estimates toward zero trades a small amount of in-sample fit for scores that will generalize more reliably to a new sample. For a study being designed around fresh data collection, that tradeoff is worth making.

I do want to acknowledge the construct validity concern. My alternative items target practices students encounter daily. If the research team cares specifically about what students experience, those items have a better argument on face validity grounds; statistical fit and construct validity are different things. I would encourage piloting both item sets in a subsample and examining whether the two indices converge before committing to one.

The Spearman correlation of 0.31 is what concerns me most. A school in the top decile under the F&S items may rank in the middle under the alternative. Targeting interventions using either index alone risks misclassifying a large share of schools; no single index should be treated as ground truth.

Region, school size, and lunch eligibility all independently predict security after Bonferroni correction; urbanicity does not. Two of four regions score meaningfully higher than the reference, suggesting that geographic location shapes security culture in ways that go beyond simple urban/rural distinctions. School size and economic disadvantage both point in the same direction: larger, poorer schools implement more intensive security measures. The research team should treat all three significant variables as covariates in any analysis linking security to student outcomes; the associations are strong enough to confound naive comparisons.

Urbanicity can be dropped without much loss.

Three limitations stand out. First, the MAR assumption is untestable; if security-conscious schools systematically skip survey items, estimates will be biased downward. Second, ELS:2002 is from 2002 and school security has changed substantially since then, particularly following high-profile incidents of school violence. Third, the Rasch model assumes all items discriminate equally well between high- and low-security schools. A 2-parameter item response model would relax this; the research team should consider whether that added flexibility is warranted given their anticipated sample size.

References

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